

Procedural Continuity and Governing-Capacity Loss in AI-Assisted Institutions

A Descriptive State Model with Pre-Abuse Collapse as a Provisional Etiological Subtype

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Abstract

AI governance can mistake procedural completion for evidence that a human-controlled institution remains intact. That inference can fail. A workflow may complete every required step while human contributions become derivative, corrective voice becomes ineffective, practical authority fragments, and a declared protected function deteriorates. This paper develops a middle-range model of procedurally masked governing-capacity loss that is designed for empirical rejection but not yet validated. It separates a descriptive state from an etiological claim. Realized procedurally masked governing-capacity loss (PMGCL) requires stable or improving visible procedural fidelity, material and persistent declines in independent judgment and effective governing capacity, and material impairment of a predeclared protected function. Pre-Abuse Collapse (PAC) is retained only as a provisional etiological subtype: PAC additionally requires an identified, materially adverse ordinary-exposure contrast under a predeclared malicious-pathway intervention or reference condition. The contrast does not prove that malicious conduct is absent, unnecessary in a sufficient-component-cause sense, or irrelevant to any individual case. If protected-function impairment is unshown, the state is incipient governing-capacity-loss risk; if the causal contrast is unidentified, PMGCL may be classified but PAC may not.

The theory integrates established component mechanisms rather than claiming to invent them. Cognitive externalization, legibility capture, responsibility inversion, and voice attenuation are specified as candidate pathways with pressures, observable traces, countervailing conditions, rival explanations, and falsifiers. A shared fifteen-unit public-record corpus from official investigations of Robodebt, the Dutch childcare-allowance scandal, and Michigan unemployment-insurance claimant services is used only to establish traceability of mechanism-adjacent objects such as evidence sufficiency, categorical simplification, review access, feedback routing, warning persistence, and remedy. It does not classify any case as PAC, estimate prevalence, identify causal effects, or validate temporal sequence.

The paper derives five rejectable propositions and a future identification program. The fifth is a comparative retirement test: binary PMGCL and PAC classifications must be compared, under nested site- or time-blocked validation, with continuous component models and established-theory baselines on an independent outcome or intervention choice. Calibration and decision value must be reported; the protected-function element used to define the state cannot be reused as the target. PMGCL or the PAC label should be retired if it adds no reproducible discrimination, lead time, calibration, or intervention value beyond simpler models.

Keywords: AI governance; governing capacity; human oversight; institutional decoupling; organizational silence; procedural fidelity; sociotechnical systems

Generative AI disclosure for SSRN abstract metadata: OpenAI ChatGPT/Codex assisted with literature discovery, adversarial review, structural and language revision, figure review, document generation, and quality assurance. An Anthropic Claude critique supplied by the author informed the v5.8 adversarial review. Version 6.2 aligns the PAC language with the estimand actually specified, rewrites the synthetic identification example to avoid post-exposure conditioning ambiguity, adds a current relational-governance comparator, and normalizes publication structure. The author remains responsible for verifying every claim, citation, analysis, and conclusion before public posting.

1. Introduction

Governance failure is easy to recognize when it has a villain or a spectacular event: an attacker exploits a model, an organization conceals known bias, or a defective system produces visible harm. The analytically harder case is an institution in which people follow approved procedures, a system remains within technical tolerances, and the workflow nevertheless loses the ability to identify, contest, correct, explain, remedy, or stop consequential decisions.

This possibility follows from a sociotechnical fact. Decisions are produced not by a model alone but by representations, targets, interfaces, queues, review roles, escalation channels, managerial responses, vendor dependencies, and remedies. Each local adaptation may be defensible. Their conjunction can still remove decision-relevant information and redistribute practical authority away from the actors nominally charged with oversight. Procedural completion can then remain stable - or improve - while the procedure's protective function decays.

This paper names the descriptive realized state procedurally masked governing-capacity loss (PMGCL). It retains Pre-Abuse Collapse (PAC) only as a provisional etiological subtype. 'Pre-abuse' does not mean that collapse must occur before abuse in calendar time, nor does it require proving that every actor behaved well. It is shorthand for a narrower causal contrast: under an explicit graph, estimand, assumptions, and design, does a predeclared ordinary organizational exposure retain a materially adverse effect under the stated malicious-pathway intervention or reference condition? If that contrast is unidentified, PMGCL may still be described, but PAC cannot be claimed. The label must not be read as proof that malicious conduct is absent or as a sufficient-component-cause statement about necessity.

The research question is: under what conditions can visible procedural fidelity mask a persistent loss of independent judgment, effective governing capacity, and protected-function performance in an AI-assisted workflow, and what additional evidence is needed to make an etiological PAC claim? The paper answers by separating four descriptive constructs from the causal subtype condition, locating the model against mature adjacent literatures, specifying a provisional and non-exhaustive set of four candidate mechanisms, deriving five propositions with predeclared rejection conditions, and defining retirement tests for both labels.

The ambition is middle-range and diagnostic. The paper does not claim a universal law of bureaucracy, a deterministic life cycle, a validated construct, or a quantitative prediction system. Its strongest current result is a disciplined theoretical specification whose target objects can be audited in independent official records.

2. Method and Claim Discipline

2.1 Problem-driven theory synthesis

This is a theory-development article with a bounded document probe, not a systematic review and not an empirical outcome study. The conceptual method follows a problem-driven synthesis: define the focal phenomenon; separate the what, how, why, and where/when of the theory; compare the conjunction with established alternatives; derive countervailing conditions, rival

explanations, and observations that would count against it; and cap every conclusion at the evidence level that supports it (Whetten, 1989; Jaakkola, 2020).

Evidence hierarchy. Peer-reviewed sources and official investigations anchor established mechanisms and public-record statements. Recent 2026 preprints and working papers are used mainly to delineate the emerging novelty boundary; they do not validate PMGCL, PAC, or any measurement instrument, and source status must be rechecked at each public release.

1. Declare the protected function, unit, and temporal interval before labeling failure.
2. Distinguish directly observable traces from latent mechanisms, state classifications, and realized outcomes.
3. Specify the ordinary exposure, malicious-pathway variable, protected outcome, population, causal estimand, graph, assumptions, support, and sensitivity analysis rather than infer etiology from claims about actor intent.
4. Compare the proposed conjunction with established component theories and the closest recent alternatives.
5. Derive moderators, rival explanations, temporal predictions, intervention implications, and falsifiers, including an incremental-validity test on outcomes not reused to define the state.
6. Prefer the simpler established explanation if the composite construct adds no discriminant or intervention value.

2.2 Public-record mechanism probe

The document probe asks a much narrower question than case validation: can an independent reader locate the proposed mechanism-adjacent objects in official records without relying on this paper's terminology? The trilogy uses fifteen passage-level evidence units from three purposively selected negative cases - Robodebt, the Dutch childcare-allowance scandal, and Michigan unemployment-insurance claimant services (Royal Commission into the Robodebt Scheme, 2023; Childcare Allowance Parliamentary Inquiry Committee, 2020; Michigan Office of the Auditor General, 2016). One author performed theory-directed coding; no inter-coder reliability is claimed.

2.3 Evidence ladder and no-claim rules

- A passage mapped to a mechanism is not evidence that the mechanism caused the case.
- Co-occurring codes are not a causal sequence.
- An official finding is not a construct-validation study.
- A negative case cannot establish sensitivity, specificity, prevalence, or lead time.
- Observing high procedural completion while one governance trace declines is not sufficient to classify PAC.
- If established theories explain the observations and intervention choices without the PAC conjunction, parsimony favors those theories.

2.4 Shared-corpus reuse and theory-first risk

The fifteen-unit corpus is also used in Parts I and III. Reuse keeps source locators stable and makes it possible to ask whether different governance objects can be traced in the same official records, but it is not three independent tests. Because all three papers were developed with

knowledge of the same purposively selected negative cases, the design is vulnerable to narrative fit and theory-first coding. A passage that is useful for a gate, a PAC-adjacent mechanism, and an exit obligation does not thereby validate any of them or create triangulation. Future PAC evaluation must use cases and comparison units selected independently of the PAC vocabulary, including successful, recovered, and boundary cases.

3. Theoretical Location and Novelty Boundary

3.1 Institutional decoupling and audit ritual

Institutional theory already explains why formal structures can confer legitimacy while operational practice remains partly decoupled from them (Meyer and Rowan, 1977). Audit regimes can intensify this gap when inspectable artifacts become proxies for an underlying public purpose (Power, 1997). PAC therefore does not claim to discover the divergence between formal compliance and substantive performance. Its narrower focus is an AI-assisted workflow in which visible procedural completion itself becomes a masking condition while independent contribution and corrective capacity decline.

3.2 Discretion, algorithmic control, and system-level bureaucracy

Street-level bureaucracy locates implementation in frontline discretion under resource constraint (Lipsky, 1980), while later work shows that administrative reforms redistribute rather than simply remove discretion (Brodkin, 2011). Digital systems can move discretion into categories, thresholds, queue priorities, and interface defaults; Bovens and Zouridis (2002) describe the shift from street-level to system-level bureaucracy. Ethnographic research likewise shows that users negotiate and resist algorithmic systems rather than passively receive them (Christin, 2017), and algorithmic control can create contested terrain around direction, evaluation, and discipline (Kellogg, Valentine, and Christin, 2020). PAC concerns the residual human stage that remains visible after much discretion has migrated elsewhere: does that stage still contribute independent case information, can it alter the action, and does the institution respond when it does?

3.3 Automation bias, cognitive integrity, and meaningful oversight

Human-factors research already documents automation-related complacency and bias and the loss of active processing under automation (Parasuraman and Manzey, 2010; Endsley and Kiris, 1995). Recent 2026 work narrows the boundary further. Lin et al. (2026) define a cognitive-integrity threshold below which oversight becomes procedural because humans cannot retain the understanding needed to verify and intervene. Gaube et al. (2026) provide an explicit architecture for effective human oversight. Van de Sande et al. (2026) identify epistemic capacity, cognitive space, decisional authority, and intervention effectiveness as conditions for meaningful medical-AI oversight. PAC cannot claim novelty for the diagnosis that nominal human presence is insufficient.

3.4 Organizational silence and corrective voice

Organizational silence research already explains why employees withhold concerns when speaking appears unsafe or futile and why managerial openness and perceived efficacy shape voice (Morrison and Milliken, 2000; Detert and Burris, 2007; Morrison, 2014, 2023). PAC uses this literature rather than renaming it. The candidate contribution is to treat the supply and efficacy of corrective voice as one component of a broader workflow-level conjunction that also includes procedural masking, judgment contribution, practical authority, and protected-function performance.

3.5 Safety drift, update opacity, and recent governance-hollowing accounts

Safety science already explains movement toward failure through locally rational adaptations under competing pressures (Rasmussen, 1997; Vaughan, 1996). Ferrario and Hatherley (2026) show how system updates can make previously acquired calibration unusable under real role and time constraints. Tallam (2026) distinguishes surviving records from the institutional practice required to interpret distributed cognitive episodes. Frimpong (2026) argues that expanding formal governance can itself weaken operational control through authority fragmentation and symbolic governance. Niu (2026) describes a related migration of effective governing capacity toward intermediary layers that control information routing and evaluative signals. Havyarimana (2026) directly distinguishes formal human authority from practical governing capacity and describes the hollowing of office when human assent remains formally decisive but substantively compelled. Simons and Broniatowski (2026) use role-conditioned stress tests to examine why the NIST AI RMF may be difficult to adopt across organizational roles; because their evidence is LLM-simulated rather than field-observed, it is adjacent evidence about role-dependent implementation frictions, not validation of PAC.

These accounts make the novelty problem severe. PMGCL earns separate descriptive use only if its pre-specified conjunction at the workflow-outcome level improves explanation, discrimination, temporal inference, or intervention choice beyond those components. PAC bears the additional burden of reproducible causal identification. The paper therefore treats novelty as an empirical burden, not a branding claim.

A further peer-reviewed comparator is Engin's Human-AI Governance (HAIG) framework, which represents decision authority, process autonomy, and accountability configuration as continuous governance dimensions with threshold changes rather than static human-in-the-loop categories (Engin, 2026). PMGCL therefore cannot claim novelty merely for tracking authority redistribution or dynamic human-AI governance. Its proposed increment remains the realized four-part conjunction of visible procedural fidelity, independent-judgment loss, practical governing-capacity loss, and protected-function impairment, together with explicit retirement conditions if that conjunction adds no independent value.

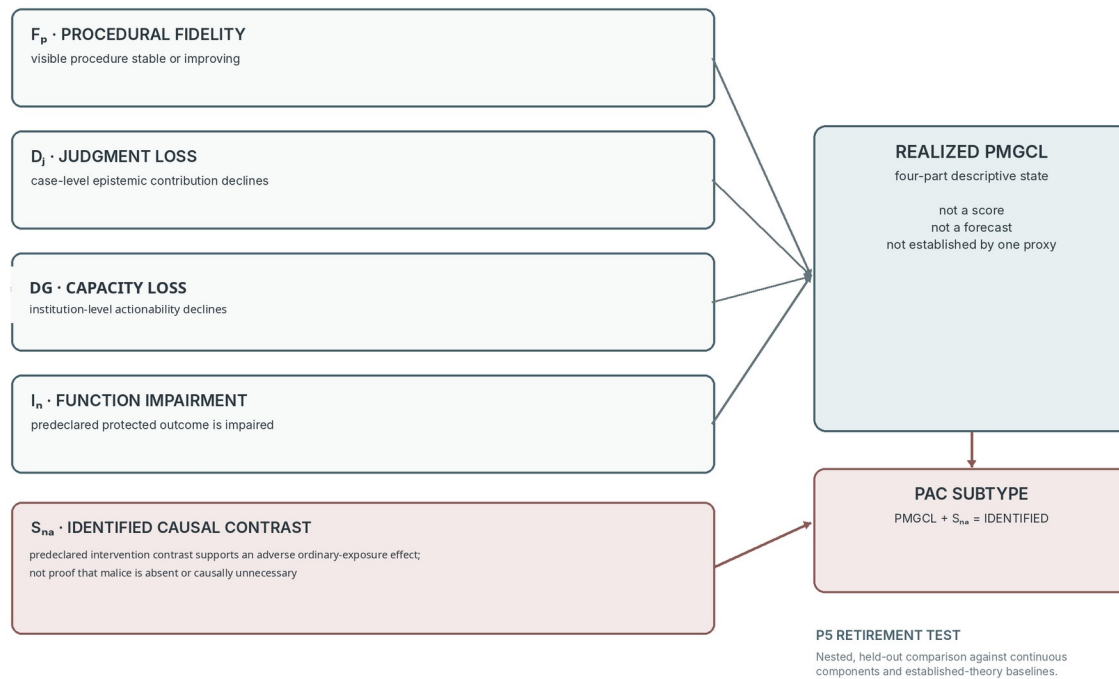
Adjacent theory	Established object	What PMGCL/PAC adds if validated	What defeats the added label
Institutional decoupling	formal structure diverges from substantive practice	pre-specified joint trajectory of procedure, judgment, capacity, and protected function	decoupling measures explain the same cases and responses

Adjacent theory	Established object	What PMGCL/PAC adds if validated	What defeats the added label
Safety drift	locally rational adaptation moves systems toward failure	stable procedural completion as masking condition plus normative outcome	no added mechanism, discrimination, or intervention relevance
Out-of-the-loop performance	automation degrades situation awareness / takeover	institution-level voice, authority, remedy, and protected function	individual human-factors measures fully explain impairment
System-level bureaucracy	discretion migrates into systems and designers	test whether the retained human stage still contributes and can correct	relevant discretion has already moved entirely upstream
Organizational silence	low safety/efficacy suppresses voice	voice becomes one component in a coupled governance-capacity state	voice theory alone predicts the impairment and remedy
Cognitive integrity / hollowing	formal authority persists while understanding or governing capacity erodes	explicit realized protected-function condition and redundancy test	new conjunction adds no out-of-sample value

Table 1. Incremental boundary relative to established and recent adjacent theories.

DESCRIPTIVE STATE FIRST; ETIOLOGICAL SUBTYPE SECOND

PMGCL is descriptive, PAC additionally requires an identified ordinary causal contrast.



J and G can dissociate: strong reasoning without power (high J / low G), or power with derivative reasoning (low J / high G).

Figure 1. PMGCL is the four-part descriptive state. PAC is a narrower provisional etiological subtype requiring a separately identified ordinary causal contrast; neither label is validated here.

4. Core Constructs, Descriptive State, and Etiological Subtype

For a declared workflow, protected function, population, and interval I , define four distinct objects. P denotes visible procedural fidelity: completion of required steps, documentation, approvals, and other observable process artifacts. J denotes case-level epistemic contribution: new evidence, reasoning, uncertainty qualification, or correction beyond the AI output and procedural template. G denotes institution-level actionability: the practical ability to make such contribution consequential by modifying action, escalating, mobilizing resources, providing remedy, or stopping the workflow. N denotes performance of the predeclared protected function - for example lawful proportional treatment, safety, due process, or another normatively specified outcome. N is not automatically model accuracy.

$$PMGCL(I) = 1 \text{ iff } F_P(I) \wedge D_J(I) \wedge D_G(I) \wedge I_N(I)$$

$$PAC(I) = 1 \text{ iff } PMGCL(I) = 1 \wedge S_NA(I) = IDENTIFIED$$

Here F_P means visible procedural fidelity is stable or improving relative to a justified baseline; D_J means independent judgment has materially and persistently declined; D_G means effective governing capacity has materially and persistently declined; and I_N means the declared protected function is materially impaired. $PMGCL$ is defined as a descriptive classification of the full four-part conjunction: it states a joint trajectory without assigning an ordinary or malicious cause. The four elements are formative and non-interchangeable; failure of any element defeats a realized $PMGCL$ classification, while satisfaction of all four establishes the classification within the declared measurement model. The same trace must not be reused mechanically across elements without an explicit measurement model and sensitivity analysis for shared error.

S_NA is an additional causal-identification condition for the PAC subtype. Let A_O denote a predeclared ordinary organizational exposure, A_M a malicious or adversarial pathway variable, Y_N the protected-function outcome coded so higher values mean better protection, and population P the declared target population. One admissible target is a controlled or interventional contrast $\Delta_O(m) = E[Y_N(1,m) - Y_N(0,m) | P]$, with the intervention/reference condition m fixed in advance when that estimand is substantively coherent and identifiable. S_NA takes one of three reporting states - IDENTIFIED, BOUNDED/PARTIALLY IDENTIFIED, or UNIDENTIFIED - and is IDENTIFIED only if the design supports a materially adverse ordinary-exposure contrast, for example $\Delta_O(m) \leq -\delta_N$ for a predeclared materiality threshold $\delta_N > 0$, under a defensible graph, consistency, positivity/support, interference, measurement, timing, and confounding assumptions. Other estimands may be used, but they must be specified before analysis and their interpretation must match the PAC claim.

This estimand does not establish individual sufficient causation, does not prove the absence of malicious conduct, and does not establish that malicious conduct is unnecessary in the sufficient-component-cause sense. A directed acyclic graph alone is not evidence. If A_M is downstream of A_O or otherwise lies on a post-exposure pathway, the contrast cannot be estimated by simply restricting or adjusting to observed $A_M=0$; doing so can create post-treatment selection or mediator-outcome confounding bias. The design must identify the intervention estimand under the relevant longitudinal assumptions and use an appropriate g-method, weighting, standardization, or other justified estimator rather than naive stratification (Hernán and Robins, 2020). Only $S_NA=IDENTIFIED$ permits

a PAC-subtype classification. Partial identification can support bounds or investigation but not the etiological label. Authors must publish the graph, estimand, assumptions, measurement timing, support diagnostics, rival paths including A_M, estimation code, and sensitivity analysis; a verbal statement that 'no malice was needed' is not evidence.

Element	Operational target	Discriminant test	Prohibited shortcut
P · procedural fidelity	visible step, approval, record, or service-level completion	can P remain high while J or G changes?	document volume as substantive quality
J · judgment contribution	case-level information or reasoning beyond AI/template	J can be high while the actor lacks power to act	override/disagreement alone
G · governing capacity	institutional ability to alter, resource, escalate, remedy, or stop	G can be high where a particular rationale is derivative	formal role or org chart alone
N · protected function	independently measured, predeclared outcome	not used again as the P5 validation target	model accuracy by default
S_NA · causal boundary	predeclared ordinary-exposure contrast meets materiality threshold under stated assumptions	rival malicious and ordinary paths estimated or bounded	absence of proven malice; DAG alone

Table 2. Construct separation. J is epistemic contribution; G is institutional actionability. Their measures may correlate but may not be reused by definition.

4.1 Incipient risk is not realized loss

If credible evidence shows declining J and G while protected-function impairment is not yet demonstrated or observable, the correct state is incipient governing-capacity-loss risk. If all four descriptive conjuncts are established but S_NA is unidentified, the correct state is realized PMGCL with etiology unresolved—not PAC. This aligns Part II with Part III's detection ontology: an indicator may open investigation, but classification requires independent substantive evidence and does not itself authorize containment or retirement.

4.2 "Pre-abuse" is an etiological boundary, not a moral chronology

Adversarial conduct can coexist with ordinary organizational causes. The theory must not suppress evidence of deception, discrimination, retaliation, or unlawful conduct to preserve the label. The causal question is whether the predeclared ordinary exposure has a materially adverse protected-outcome contrast under the stated malicious-pathway intervention or reference condition and declared assumptions. If the contrast is not identified, or if it is smaller than δ_N , the case is not evidence for the PAC subtype even if PMGCL is descriptively present. The label therefore says nothing by itself about the existence, moral status, or individual causal necessity of malicious conduct. In particular, PAC must not be assigned by conditioning a retrospective dataset on 'no documented malice' when that status can itself be affected by exposure, detection, reporting, or institutional response.

4.3 Synthetic identification example

Suppose a benefits-review organization staggers a queue-cap policy that cuts protected review minutes, A_O, across otherwise comparable teams for operational reasons unrelated to the study. Before rollout, investigators register P as completion of required review steps, J as blinded

independent-coder evidence contribution, G as authority-weighted successful correction capacity, and Y_N as independently reconstructed lawful-outcome quality. They set $\delta_N=0.05$ and register a causal graph that places the malicious-pathway variable A_M explicitly in time. If A_M is fixed upstream of A_O , a controlled contrast at a predeclared reference level may be estimable under the stated assumptions. If A_M is affected by A_O , investigators instead define an interventional or longitudinal estimand and use a justified g -formula, inverse-probability weighting, standardization, or another estimator appropriate to the graph; they do not restrict the observed sample to $A_M=0$. A staggered event-study or other quasi-experimental component may identify the A_O variation only if timing, parallel-trend or exchangeability conditions, support, and negative-control diagnostics are defensible. PAC is eligible only if the registered ordinary-exposure contrast is materially adverse - for example below -0.05 with uncertainty excluding immaterial effects - and the malicious-pathway handling is itself identified. If timing is endogenous, overlap fails, A_M is poorly observed, the longitudinal assumptions fail, or the interval includes zero or effects smaller than δ_N , S_{NA} remains bounded or unidentified. This is a study template, not evidence that any real case satisfies PMGCL or PAC.

4.4 Protected functions must be predeclared and plural

A workflow can improve one outcome while damaging another. Faster throughput can coexist with worse proportionality; lower model error can coexist with inaccessible remedy. A PAC study must therefore predeclare the protected function and its measurement. Treating every desirable outcome as one composite 'governance quality' variable would make the theory unfalsifiable.

5. Four Candidate Mechanisms

These four mechanisms are a provisional, theory-directed set, not a sufficient or exhaustive taxonomy. Deskilling, vendor lock-in, update opacity, automation-induced dependency, resource withdrawal, or other mechanisms may produce the same governing-capacity pattern and should be added, substituted, or preferred when they improve identification. PMGCL concerns the pre-specified descriptive conjunction, while any PAC-subtype claim additionally depends on a predeclared causal contrast; neither label owns a closed list of pathways.

5.1 Cognitive externalization

Pressure: throughput demand, task complexity, repeated exposure, and interface design that presents the model output as the primary case representation. Mechanism: reviewers increasingly treat the model output as both the starting and ending representation of the case, reallocating attention away from independent evidence search. Expected traces include a declining complexity-responsive review-time slope, rationales that add less held-out information beyond the model and declared controls, and intervention concentrated on obvious flags. Countervailing conditions include independent evidence views, diagnostic sampling, protected time, rotation, and feedback on missed errors. Externalization is not inherently harmful; offloading can improve performance. The failure condition is loss of complementary scrutiny while the institution continues to count the human stage as independent oversight.

5.2 Legibility capture

Pressure: audit and performance systems reward what can be counted quickly. Mechanism: actors optimize completion, agreement, documentation length, queue clearance, or other legible artifacts because those artifacts are easier to manage than the protected function. Expected traces include rising procedural completion while rationale contribution, remedy timeliness, case reconstruction, or risk-adjusted protected outcomes worsen; templates may become longer while less discriminative. Countervailing conditions include outcome-linked audit, randomly sampled case reconstruction, plural metrics, and explicit anti-gaming review. Legibility capture does not require fraud; locally rational compliance with the measured target can be sufficient.

5.3 Responsibility inversion

Pressure: the reviewer is held responsible for delay or deviation but lacks authority over model design, staffing, policy, vendor constraints, or fallback. Mechanism: formal accountability concentrates on the actor least able to change the system. Expected personal cost favors acceptance, while upstream owners can treat the human signature as risk transfer. Expected traces include high approval with extensive attestations, overrides that require exceptional justification, and incidents framed as operator error despite structural causes. Countervailing conditions include authority-responsibility parity, explicit upstream ownership, protected override, and escalation to a competent independent decision maker. This differs from using moral crumple zones as a post-failure blame concept: the candidate mechanism concerns how anticipated liability can shape behavior before the incident.

5.4 Voice attenuation

Pressure: concerns are ignored, punished, routed back to the same owner whose system is challenged, or repeatedly fail to produce remedy. Mechanism: perceived efficacy of speaking up declines; the future supply of dissent then contracts, depriving the institution of corrective information. Expected traces include prior unremedied dissent predicting later decline in risk-adjusted dissent, genericized reports, and recurrence without systemic response. Countervailing conditions include protected reporting, visible remedy, independent recipients, and feedback to reporters.

VOICE ATTENUATION AS A FEEDBACK HYPOTHESIS

Unremedied voice can reduce the expected value - and later supply - of corrective voice.

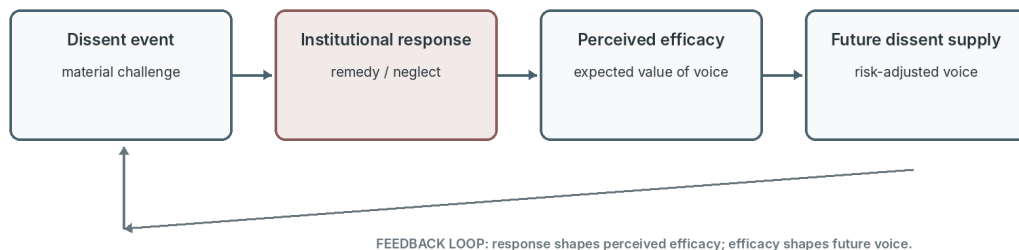


Figure 2. Voice attenuation is a feedback hypothesis; raw dissent counts alone are not evidence of governance quality.

6. Public-Record Traceability Illustrations

The shared corpus does not classify Robodebt, the Dutch childcare-allowance scandal, or Michigan claimant services as PMGCL or PAC. PMGCL classification would require predeclared longitudinal measures of P, J, G, and N; PAC would additionally require the pre-specified causal graph, estimand, assumptions, and design described above. The official records do not provide that research design. The analysis instead asks whether mechanism-adjacent objects are independently auditable.

Candidate mechanism	Robodebt	Dutch inquiry	Michigan audit	Inference permitted
Cognitive externalization	recommendations emphasize legal foundations, review, explanation, scrutiny; cognition not directly measured	mass processing and limited individual consideration	sampled determinations show fact-gathering / reason deficiencies	evidence handling and reasons are observable; cognitive offloading is not established
Legibility capture	decision-record and legal-basis recommendations show missing governance objects; target conversion not observed	group-based, risk-selected, all-or-nothing administration makes categorical simplification salient	process quality findings, not an explicit target effect	categorical simplification is traceable; target conversion remains untested
Responsibility inversion	responsibility and legal-accountability reforms show importance of locating authority	responsibility assigned across administration, legislature, judiciary; workflow authority not reconstructed	agency process and responses, not reviewer-level authority map	current corpus is insufficient to establish authority-responsibility inversion
Voice attenuation	frontline consultation and management response appear as remedial objects	persistent warning signals and delayed recognition/remedy	feedback was not systematically compiled and practical communication was weak	signal routing and response are observable; the feedback mechanism remains untested

Table 3. Public-record traceability of mechanism-adjacent objects and the inferences not supported.

The probe changes the theory in three ways. It confirms that evidence, reasons, practical review, feedback routing, managerial response, and remedy can be source-located outside the conceptual model. It forces nominal review access to be separated from practical review capacity. And it exposes a weak point: role-level effective authority and responsibility inversion are poorly recoverable from the fifteen coded units. A future test therefore needs an authority map and longitudinal workflow data, not merely policy documents or post-incident recommendations.

7. Derived Propositions

Proposition	Unit / temporal contrast	Target estimand	Evidence against
P1 Capacity-strain decoupling	reviewer/team before vs after exogenous throughput increase with staffing held fixed	change in J relative to change in P	J stable/rising or decline explained by case mix, model improvement, or learning

Proposition	Unit / temporal contrast	Target estimand	Evidence against
P2 Target conversion	workflow before/after completion or agreement target vs unaffected unit	target effect on visible artifact and conditional relation to evidence/protected outcome	both procedure and protected function improve with no burden displacement
P3 Voice-efficacy feedback	team-window sequence; institutional response precedes later voice	effect of prior remedy efficacy on later risk-adjusted dissent	no temporal association or relation explained by lower dissent opportunity
P4 Coupled-trace discrimination	held-out windows reconstructed independently	incremental discrimination of joint rationale, time, and voice traces	single raw metrics perform equally well or coupled signal fails under benign efficiency
P5 Comparative state validity	nested held-out sites/windows; outcome not used in state definition	calibrated and decision-relevant gain over continuous components and established baselines	gain vanishes, reflects leakage/dichotomization, or simpler model chooses the same intervention

Table 4. Derived propositions, target contrasts, estimands, and rejection conditions.

7.1 P1 - Capacity-strain decoupling

Within comparable units, an exogenous increase in throughput pressure without proportional capacity should produce a larger decline in independent judgment contribution than in procedural completion. Staffing increases, protected review time, or randomized queue relief should moderate the effect. The proposition is defeated if the apparent decoupling disappears after controls for case difficulty, model quality, reviewer learning, policy, and interface change.

7.2 P2 - Target conversion

Introducing a salient completion or agreement target should improve the targeted artifact while reducing its conditional relationship to case evidence or the protected outcome if legibility capture is operating. A strong design uses staggered target introduction, unaffected comparison units, and negative-control metrics. The proposition is defeated if the target improves both the artifact and protected function without hidden burden transfer.

7.3 P3 - Voice-efficacy feedback

Lower remedy efficacy for substantiated dissent at time t should predict lower risk-adjusted dissent at $t+1$, controlling for opportunities to dissent. Risk adjustment is essential: lower raw dissent can reflect a better model or easier case mix. Mediation through perceived efficacy can be tested with confidential repeated measures, but behavioral traces should remain primary where survey response itself may be chilled.

7.4 P4 - Coupled-trace discrimination

A joint signal combining rationale-information loss, time-complexity decoupling, and depletion of corrective voice should distinguish independently reconstructed governance impairment from benign efficiency better than any raw indicator alone. The proposition is defeated if the combined signal is unstable across sites, adds no held-out information, or relies on a proxy whose meaning changes under intervention. Part III's candidate GDIs operationalize this proposition but do not validate it.

7.5 P5 - Comparative state and decision validity

In held-out sites or time blocks, binary PMGCL and PAC classifications should be compared with: (i) preregistered continuous P/J/G/N component models; (ii) the same components without a conjunction; and (iii) established-theory baselines for institutional decoupling, safety drift, organizational silence, cognitive integrity, or another domain-appropriate account. The target must be independent of the state definition—for example persistence after a standardized corrective action, recovery time, recurrence, or which intervention restores the protected function. Contemporaneous N impairment cannot be reused as the target. Hyperparameters, thresholds, feature selection, and missing-data choices belong inside nested cross-validation with site- or time-grouped outer folds. Report calibration, uncertainty, subgroup performance, and decision-curve or explicit utility analysis, not only discrimination. The proposition is defeated if an apparent gain disappears after leakage and optimism controls, is merely an artifact of dichotomization or class prevalence, fails across sites, or does not alter a defensible decision.

8. Identification and Falsification Strategy

8.1 Measurement model before causal model

Every construct requires a content-valid operational definition before causal analysis. Procedural fidelity is not document length. Independent judgment is not disagreement alone. Governing capacity is not a formal org chart. Protected-function performance is not model accuracy unless the protected function is explicitly accuracy. J and G require separate measures or an explicit joint model: an analytically strong reviewer can have no practical power (high J, low G), while a powerful stage can rubber-stamp derivative reasoning (low J, high G). Researchers should document construct-measure links, cross-loadings, shared-method error, known blind spots, latency, subgroup validity, and expected behavior under benign alternatives (Jacobs and Wallach, 2021).

Minimum J/G protocol. At least two trained coders, blinded to protected outcomes and the proposed state label, independently code a stratified case sample using a published codebook. J is reconstructed from source evidence and rationale contribution beyond the model/template; G is reconstructed separately from a versioned authority map, observed escalation/correction attempts, response latency, and whether the actor could alter action or mobilize remedy. Report raw agreement, an appropriate chance-corrected or reliability coefficient with uncertainty, adjudication rules, missingness, and sensitivity to alternate thresholds. A held-out auditor should then attempt outcome and decision reconstruction from the record. Formal role titles, override counts, or text length cannot substitute for this protocol.

8.2 Temporal and counterfactual identification

A credible PMGCL study needs temporal variation that is plausibly upstream of the descriptive sequence—for example staffing shocks, target changes, interface changes, provider updates, or changes in escalation structure—plus independently measured protected outcomes. A PAC-subtype study must go further by identifying the registered ordinary-exposure contrast while measuring or bounding malicious pathways. Interrupted time-series, difference-in-differences,

event-study, stepped implementation, randomized queue relief, or carefully reconstructed matched comparisons may be appropriate depending on domain. A post hoc chart of a falling proxy and a bad outcome is not sufficient.

8.3 Alternative explanations

- Benign efficiency: expertise or interface improvement reduces review time while judgment quality remains intact.
- Model improvement: lower dissent or override is appropriate because the model became more accurate.
- Case-mix shift: simpler cases alter time, rationale, and dissent without governance degradation.
- Policy or legal change: the target function or permissible actions legitimately changed.
- Measurement displacement: workers move substantive reasoning to an unobserved channel rather than ceasing to reason.
- Outcome-selection and delay: appeals, verified errors, and harm labels arrive selectively or too late for the proposed sequence.
- Adversarial or discriminatory conduct: malicious or unlawful behavior is a material cause and must be modeled rather than excluded rhetorically.

8.4 What would falsify or retire the construct?

PMGCL should be retired as a separate construct if the four-part conjunction adds no reproducible explanation, calibration, temporal inference, or intervention value beyond preregistered continuous components and established theories; if J and G cannot be discriminated reliably; or if protected-function measurement cannot support a realized state. The PAC subtype should additionally be retired if S_NA cannot be operationalized reproducibly or causal estimates do not survive alternate graphs, bounds, negative controls, and sensitivity analyses. Any apparent gain that reuses N or another constitutive measure as the validation target, leaks site/time information, or vanishes under nested validation is disqualifying. A named theory should not survive merely because it is memorable.

9. Governance Implications

The theory shifts audit questions from 'Was a human present?' to four linked questions: Did the human contribute information not already contained in the model or template? Could they act on that information? Did the institution respond when corrective voice was supplied? Did the predeclared protected function remain intact? Completion and agreement remain useful process measures, but their relationship to evidence and outcomes must be tested rather than assumed.

It also changes accountability. If throughput, target design, vendor dependency, and authority allocation cause judgment to collapse, retraining individual reviewers is unlikely to be sufficient. Corrective action may require queue relief, interface redesign, independent escalation, changed incentives, stronger fallback, or suspension. Leadership owns those conditions.

Documentation should preserve distinctions rather than merely add text. A governance record should distinguish model output, source evidence, human contribution, unresolved uncertainty, institutional action, authority, later challenge, and correction. More documentation is not necessarily more governing capacity (Winecoff and Bogen, 2025).

10. Relationship to Parts I and III

Part I provides the case-level front door: a two-stage typed contract separates PASS, REVIEW, BLOCK, and UNKNOWN from routing status and an actual execution or signed disposition event. Part II asks whether that visible architecture can remain procedurally intact while independent contribution, corrective capacity, and protected-function performance deteriorate. Part III then separates weak signal, candidate event, incident, realized PMGCL, the narrower PAC subtype where identified, Resolution Collapse, containment, Proper Ending, Authority Return, closure, temporary release, and post-closure redeployment.

State / concept	Part II role	Transition to Part III
Incipient capacity-loss risk	credible J/G impairment; N impairment not shown or not observable	may justify independent investigation; not automatic containment
Realized PMGCL	four-part descriptive conjunction including protected-function impairment	may support an incident finding; does not itself require retirement
PAC subtype	PMGCL plus identified ordinary causal contrast	provisional etiological subtype: PMGCL plus an identified materially adverse ordinary-exposure contrast under a predeclared malicious-pathway intervention/reference condition; not proof that malice is absent or causally unnecessary in a sufficient-component sense
Resolution Collapse	not required for PMGCL/PAC; remediation cannot restore operation within approved horizon	one sufficient retirement basis for Proper Ending
Proper Ending / Authority Return	outside Part II theory	Part III institutional exit and disposition of decision capacity

Table 5. Part II state concepts and their transition boundary to Part III.

This separation is central to the trilogy's governance logic: diagnosis must not acquire decision authority by definition. A PMGCL or PAC classification can change the evidentiary state of the institution, but only a competent authority under the applicable process may contain, retire, resume, or redeploy the workflow.

11. Limitations and Research Ethics

PMGCL is a proposed formative descriptive construct, not a new general theory of organizational drift or a validated latent scale; PAC is a narrower provisional etiological subtype. Both deliberately overlap with institutional decoupling, safety drift, out-of-the-loop performance, system-level bureaucracy, audit ritual, automation bias, organizational silence, cognitive integrity, update opacity, and recent governing-capacity accounts. Their value depends on

incremental explanatory, discriminant, temporal, causal, and intervention validity under leakage-free held-out tests.

Designed for empirical rejection describes the architecture of the claims, not a completed empirical achievement. Every descriptive conjunct contains domain judgments about materiality, persistence, measurement, and baseline choice; S_NA additionally requires a defensible causal identification strategy. Until independent teams demonstrate reproducible operationalization, the paper supplies a rejection-oriented research program rather than a validated instrument or causal theory.

The public-record probe is narrow, purposive, author-coded, and selected for documented failure and mechanism coverage. It includes no matched successful workflow and does not independently reconstruct the investigations. It cannot establish the latent mechanisms, their sequence, the PMGCL conjunction, or the PAC causal subtype.

The same fifteen units are reused across the trilogy. That controlled reuse aids source consistency but cannot produce independent replication or cumulative validation and may amplify narrative-fit bias across all three papers.

The theory can be abused as a rhetorical label for any disliked institution. Protected functions are plural and politically contested; one workflow can improve one function while worsening another. Predeclared outcomes, rival explanations, explicit falsifiers, and the construct-retirement rule are intended to constrain opportunistic use.

Instrumentation can itself create governance harm. Review-time telemetry, rationale analysis, and dissent monitoring can become worker surveillance, speed pressure, or a new target to game. Data access should be proportionate, aggregated where possible, protected from punitive secondary use, and co-designed with affected workers and communities.

Finally, the label 'Pre-Abuse' may distract from its counterfactual meaning even when carefully defined. PMGCL is therefore the primary descriptive term. Researchers should omit PAC unless the additional causal burden is met and the label adds decision-relevant value.

12. Conclusion

An AI-assisted institution can satisfy procedure while losing the judgment, corrective voice, and effective authority that made the procedure protective. The revised model distinguishes incipient risk, realized descriptive loss, and etiological subtype. Incipient governing-capacity-loss risk requires credible evidence of declining independent contribution and governing capacity. Realized PMGCL additionally requires material impairment of a predeclared protected function while visible procedure remains stable or improves. PAC is narrower still: it requires an adequately identified, materially adverse ordinary-exposure contrast under a pre-specified causal model; a graph or absence of proven malice is insufficient.

The public-record exercise does not prove that state. It shows only that evidence sufficiency, reasons, practical review, categorical simplification, feedback routing, warning persistence, and remedy are auditable objects outside the conceptual model, while role-level effective authority remains inadequately captured. That negative finding is useful: a future study needs

longitudinal workflow data, independent outcome reconstruction, and an authority map, not a post hoc bundle of dashboard correlations.

The labels are therefore offered with separate retirement conditions. PMGCL earns separate use only if its four-part conjunction improves leakage-free explanation, calibration, timing, or intervention choice beyond continuous components and established theories on held-out data. PAC must also survive reproducible causal identification and add value beyond the descriptive state. Until those tests are met, PMGCL is a provisional state specification and PAC a provisional etiological subtype, neither a forecast nor a validated diagnosis. Part III converts that boundary into a constrained detection and accountable-exit architecture.

Appendix A. Trilogy Terminology Map

The same terms are used at different levels of analysis across the trilogy. This map prevents a diagnostic signal or preliminary state from acquiring decision authority by terminology alone.

Term	Primary level	Meaning and authority boundary
Gate state	case / Part I	PASS, REVIEW, BLOCK, or UNKNOWN; a diagnostic output, never a final institutional act
Routing status	case / Part I	Stage-1 destination such as EXECUTION ELIGIBLE, EVIDENCE HOLD, AUTHORIZED REVIEW, FALLBACK, or UNRESOLVED; never an act by itself
Execution event	case / Part I	separately authorized downstream act after EXECUTION ELIGIBLE; scope, freshness, revocation, and audit rules come from the domain contract
Signed disposition	case / Part I	actual current, in-scope Stage-2 instruction; only signed REJECT can create FINAL STOP
Incipient capacity-loss risk	institution / Part II	credible judgment/capacity decline without the full realized-state evidence
PMGCL	institution / Part II	descriptive four-part state: stable visible procedure with persistent judgment loss, governing-capacity loss, and protected-function impairment
PAC subtype	institution / Part II	provisional etiological subtype: PMGCL plus an identified materially adverse ordinary-exposure contrast under a predeclared malicious-pathway intervention/reference condition; not proof that malice is absent or causally unnecessary in a sufficient-component sense
GDI	monitoring / Part III	defeasible governance trace that may route investigation but cannot authorize action
Circuit Breaker	transition / Part III	authorized, scoped, reversible containment; not retirement or closure
Resolution Collapse	transition / Part III	authorized finding that approved remediation cannot restore operation within the declared horizon
Proper Ending	institution / Part III	evidence- and remedy-preserving retirement process; not merely switching off a model
Authority Return	institution / Part III	reconstitution, reassignment, or residual custody of decision capacity and unresolved obligations

Term	Primary level	Meaning and authority boundary
Validity hold	measurement / Part III	preserves a signal for investigation while blocking automated escalation until the measurement basis is repaired

Table A1. Trilogy terminology map and authority boundaries.

Data, Materials, and Reproducibility

The empirical material in this paper is limited to the shared trilogy public-record corpus: fifteen author-coded evidence units from three official investigations. The matrix is a traceability artifact, not validation data. A future validation package should preserve preregistered construct definitions, coder codebooks and reliability outputs, field availability and latency, protected-function measures, versioned authority maps, causal graph and estimand, analysis code, and independently sampled successful, recovered, failure, and boundary comparisons. No participant, employer, client, or confidential data are analyzed here.

Generative AI Use Disclosure

OpenAI ChatGPT/Codex assisted with literature discovery, adversarial review, structural and language revision, figure and document generation, and quality assurance. An Anthropic Claude critique supplied by the author was used as an additional adversarial-review input; its combinatorial, bibliographic, and substantive claims were independently checked before adoption. Version 6.2 aligns the PAC interpretation with the stated causal estimand, removes a residual necessary-cause overclaim, rewrites the synthetic identification example to make post-exposure handling explicit, adds a current relational-governance comparator, and normalizes table/caption structure. It does not claim empirical identification or validation. No participant or field data were supplied. The author retains intellectual and editorial control and is responsible for the claims, citations, analyses, and conclusions in the public version.

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References

- Bovens, M., and Zouridis, S. (2002). From street-level to system-level bureaucracies: How information and communication technology is transforming administrative discretion and constitutional control. *Public Administration Review*, 62(2), 174-184. doi:10.1111/0033-3352.00168
- Brodtkin, E. Z. (2011). Policy work: Street-level organizations under new managerialism. *Journal of Public Administration Research and Theory*, 21(Suppl. 2), i253-i277. doi:10.1093/jopart/muq093
- Childcare Allowance Parliamentary Inquiry Committee. (2020). Ongekend onrecht: Verslag Parlementaire ondervragingscommissie Kinderopvangtoeslag. House of Representatives of the States General of the Netherlands. Official report.
- Christin, A. (2017). Algorithms in practice: Comparing web journalism and criminal justice. *Big Data & Society*, 4(2). doi:10.1177/2053951717718855
- Detert, J. R., and Burris, E. R. (2007). Leadership behavior and employee voice: Is the door really open? *Academy of Management Journal*, 50(4), 869-884. doi:10.5465/amj.2007.26279183
- Endsley, M. R., and Kiris, E. O. (1995). The out-of-the-loop performance problem and level of control in automation. *Human Factors*, 37(2), 381-394. doi:10.1518/001872095779064555

- Engin, Z. (2026). Human-AI governance (HAIG): A trust-utility approach. *Journal of Responsible Technology*, 26, 100167. doi:10.1016/j.jrt.2026.100167
- Ferrario, A., and Hatherley, J. (2026). Update opacity: Epistemic accessibility and governance under AI system change. arXiv preprint arXiv:2606.00037. doi:10.48550/arXiv.2606.00037
- Frimpong, V. (2026). The Governance Inversion Hypothesis: Why more AI regulation may produce less organisational control (version 2). arXiv preprint arXiv:2606.26117. doi:10.48550/arXiv.2606.26117
- Gaube, S., Langer, M., Miller, T., et al. (2026). Keeping an eye on AI: A framework for effective human oversight of AI systems. arXiv preprint arXiv:2605.16278. doi:10.48550/arXiv.2605.16278
- Havyarimana, J. C. (2026). The hollowing of office: Human sovereignty, compelled assent, and the loss of discernment when AI governs in practice. SSRN Working Paper 7218298. doi:10.2139/ssrn.7218298
- Hernán, M. A., and Robins, J. M. (2020). *Causal Inference: What If*. Boca Raton: Chapman & Hall/CRC.
- Jaakkola, E. (2020). Designing conceptual articles: Four approaches. *AMS Review*, 10, 18-26. doi:10.1007/s13162-020-00161-0
- Jacobs, A. Z., and Wallach, H. (2021). Measurement and fairness. *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 375-385. doi:10.1145/3442188.3445901
- Kellogg, K. C., Valentine, M. A., and Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals*, 14(1), 366-410. doi:10.5465/annals.2018.0174
- Lin, F., Ge, Q., Xu, L., Li, P., Gao, X., Xing, S., Yamada, K., Zhang, Z., Zhang, H., and Tu, Z. (2026). Position: Human-centric AI requires a minimum viable level of human understanding. arXiv preprint arXiv:2602.00854. doi:10.48550/arXiv.2602.00854
- Lipsky, M. (1980). *Street-Level Bureaucracy: Dilemmas of the Individual in Public Services*. Russell Sage Foundation.
- Meyer, J. W., and Rowan, B. (1977). Institutionalized organizations: Formal structure as myth and ceremony. *American Journal of Sociology*, 83(2), 340-363. doi:10.1086/226550
- Michigan Office of the Auditor General. (2016). *Claimant Services: Unemployment Insurance Agency (Performance Audit Report 641-0318-14)*. Official report.
- Morrison, E. W. (2014). Employee voice and silence. *Annual Review of Organizational Psychology and Organizational Behavior*, 1, 173-197. doi:10.1146/annurev-orgpsych-031413-091328
- Morrison, E. W. (2023). Employee voice and silence: Taking stock a decade later. *Annual Review of Organizational Psychology and Organizational Behavior*, 10, 79-107. doi:10.1146/annurev-orgpsych-120920-054654
- Morrison, E. W., and Milliken, F. J. (2000). Organizational silence: A barrier to change and development in a pluralistic world. *Academy of Management Review*, 25(4), 706-725. doi:10.5465/amr.2000.3707697
- Niu, X. (2026). The chancellor trap: Administrative mediation and the hollowing of sovereignty in the algorithmic age. arXiv preprint arXiv:2602.18474. doi:10.48550/arXiv.2602.18474
- Parasuraman, R., and Manzey, D. H. (2010). Complacency and bias in human use of automation: An attentional integration. *Human Factors*, 52(3), 381-410. doi:10.1177/0018720810376055
- Power, M. (1997). *The Audit Society: Rituals of Verification*. Oxford University Press.
- Rasmussen, J. (1997). Risk management in a dynamic society: A modelling problem. *Safety Science*, 27(2-3), 183-213. doi:10.1016/S0925-7535(97)00052-0
- Royal Commission into the Robodebt Scheme. (2023). *Report of the Royal Commission into the Robodebt Scheme*. Australian Government. Official report and recommendations.
- Sato, T. (2026a). *Workflow-Centric AI Governance: A Typed Gate Contract for Accountable Human-AI Decisions*. Working paper v6.2, SSRN DOI 10.2139/ssrn.5911063.
- Sato, T. (2026b). *From Governance Drift to Accountable Exit: Proper Ending and Authority Return in AI-Assisted Institutions*. Working paper v6.2, SSRN DOI 10.2139/ssrn.6066430.
- Simons, J. R., and Broniatowski, D. A. (2026). Why AI governance frameworks are hard to adopt: A role-based stress test of the NIST AI RMF. arXiv preprint arXiv:2608.12352. doi:10.48550/arXiv.2608.12352
- Tallam, K. (2026). The khipu problem: Institutional legibility under distributed cognition. arXiv preprint arXiv:2606.12414. doi:10.48550/arXiv.2606.12414
- van de Sande, D., Economou-Zavlanos, N., and van Genderen, M. E. (2026). Meaningful oversight of medical AI beyond human in the loop. *npj Digital Medicine*, 9, 569. doi:10.1038/s41746-026-02971-1
- Vaughan, D. (1996). *The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA*. University of Chicago Press.
- Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of Management Review*, 14(4), 490-495. doi:10.5465/amr.1989.4308371
- Winecoff, A. A., and Bogen, M. (2025). Improving governance outcomes through AI documentation: Bridging theory and practice. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, Article 1194, 1-18. doi:10.1145/3706598.3713814